MIMO Antenna with FSS

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***Abstract*—This paper presents a four-port multiple-input multiple-output (MIMO) antenna placed over a frequency selective surface (FSS). Each port feeds a ten-sided (decagonal) patch through a coplanar-waveguide (CPW) style feed line. The FSS is a 6 × 5 array of unit cells, and each cell is a pair of concentric split rings. The structure is modelled and simulated in CST Studio Suite. The simulated reflection coefficient at port 1 is below −10 dB from 1.71 to 5.48 GHz and from 5.73 GHz to the end of the simulated range at 15 GHz. Between these two bands it rises to −9.2 dB at 5.62 GHz. The best match is −34.9 dB at 7.38 GHz. Port isolation, gain, envelope correlation coefficient and radiation patterns: TO BE ADDED.**

Keywords*—MIMO antenna, frequency selective surface, split-ring resonator, CPW feed, wideband antenna

I. Introduction

A MIMO system uses several antennas at each end of a wireless link. It can carry more data than a single-antenna link without using more bandwidth or more transmit power [1], [2]. For this to work, the antennas must be well matched and must stay independent of each other, so a MIMO antenna is judged by its port isolation and its envelope correlation coefficient (ECC) as well as by its bandwidth and gain [3], [4].

Wideband systems, such as ultra-wideband (UWB) links in the 3.1–10.6 GHz band [5], need antennas that stay matched over several gigahertz. Printed planar monopoles are a common choice, because a flat patch behaves like a thick monopole and gives a very wide impedance bandwidth [6], [7]. A CPW feed keeps the feed line and the ground on the same side of the board, which makes the antenna easy to fabricate [8]. The weak point of a printed monopole is its gain. It radiates to the front and to the back of the board, so about half of the power goes in the unwanted direction [9].

An FSS is a flat periodic array of metal cells that reflects or passes a wave depending on its frequency [10]. Placed behind an antenna, it sends the backward wave to the front and raises the gain. Many single-port wideband antennas use an FSS in this way [11]. Ranga *et al.* used a dual-layer FSS reflector behind a UWB antenna [12]. Kundu *et al.* added two FSS layers to a printed UWB antenna and reached a peak gain of 8.7 dBi [13]. With single-layer FSS reflectors, peak gains of 11.5 dBi [14], 10.5 dBi [15] and 8.4 dBi [16] have been reported for CPW-fed monopoles. A reflector made of split-ring cells was used behind a UWB monopole in [17].

Fewer works combine a periodic surface with a MIMO antenna. A four-port antenna over a metasurface of split-ring cells reached 8.3 dBi with more than 15.5 dB isolation in [18], and a two-port UWB antenna with FSS layers was reported in [19]. In this paper, a four-port MIMO antenna with decagonal patches is placed over an FSS made of concentric split rings. Section II describes the structure, Section III gives the simulated result that is available so far, and Section IV concludes the paper.

II. Antenna Design

A. Four-Port MIMO Antenna

Fig. 1 shows the four-port MIMO antenna. It has four radiating elements arranged in two rows and two columns. Each element is a ten-sided (decagonal) patch joined to a straight feed line. A rectangular ground patch lies on each side of the feed line, on the same face of the board, as in a CPW feed. The two upper elements are fed from the top edge of the board and the two lower elements from the bottom edge, so the upper pair is a mirror image of the lower pair and the four patches face each other at the middle of the board. The upper and lower halves are shifted sideways by a small amount, which gives the board a stepped outline. The dimensions of the antenna are listed in Table I. Substrate and port numbering: TO BE ADDED.

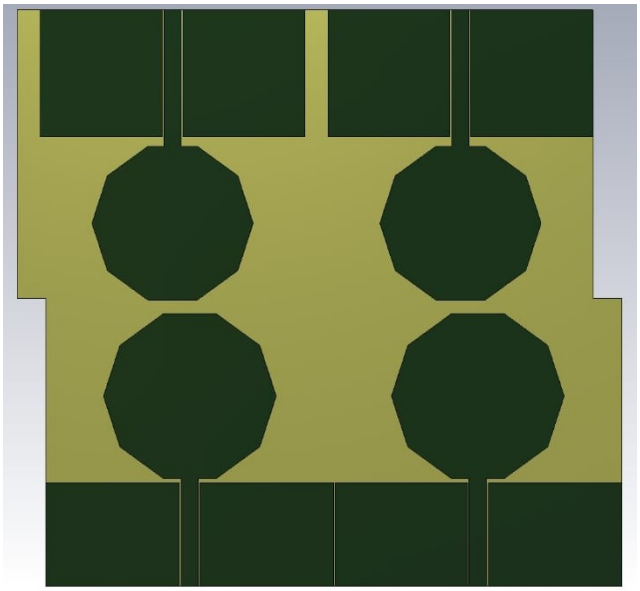


Fig. 1. Four-port MIMO antenna with decagonal patches and CPW-style feeds (top view, CST).

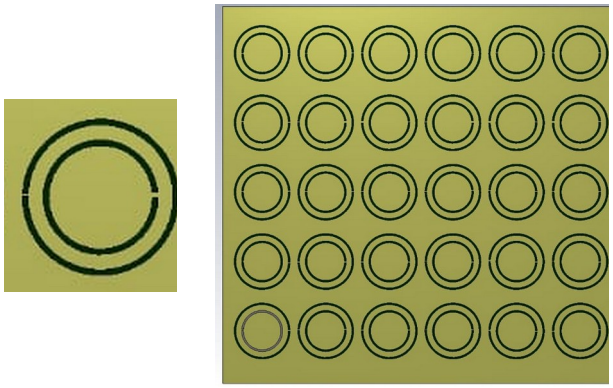
B. FSS Unit Cell and Array

The FSS is shown in Fig. 2. Its unit cell has two concentric metal rings, and each ring is cut by a narrow split. The cell is repeated in 6 columns and 5 rows, which gives 30 cells on one board. A split ring acts as a resonant circuit: the metal ring

gives an inductance L, and the split and the gap between the two rings give a capacitance C. The cell resonates at

$$f_0 = \frac{1}{2\pi\sqrt{LC}} \quad (1)$$

Near this frequency the surface reflects most of the wave that falls on it [10]. Two rings of different size resonate at two different frequencies, which widens the range over which the surface reflects. Unit-cell simulation (reflection and transmission of the FSS, and its reflection phase): TO BE ADDED.



(a) (b)

Fig. 2. FSS: (a) unit cell with two concentric split rings, (b) 6 × 5 array (top view, CST).

C. Combined Structure

In the combined structure of Fig. 3, the MIMO antenna is placed over the FSS, so the FSS lies behind all four elements. The two boards have nearly the same footprint. The wave that the antenna radiates backwards travels across the gap h, is reflected by the FSS and returns to the antenna. It adds to the forward wave when [15]

$$\varphi_{FSS} - 2k_0 h = 2n\pi, n = 0, 1, 2, \ldots \quad (2)$$

where φ_{FSS} is the reflection phase of the FSS and k_0 is the free-space wavenumber. The gap h therefore decides at which frequencies the FSS helps. Gap h between the antenna and the FSS: TO BE ADDED.

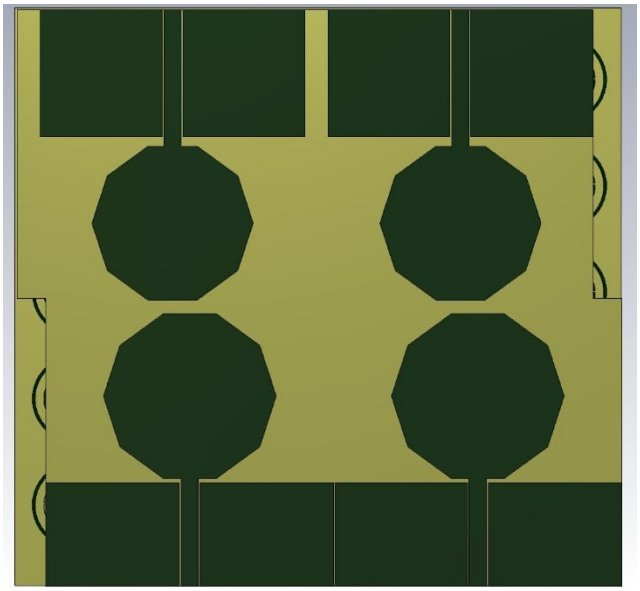


Fig. 3. Four-port MIMO antenna placed over the FSS (top view, CST).

TABLE I
DIMENSIONS OF THE PROPOSED STRUCTURE

Part	Parameter	Value (mm)
Board	Substrate material and thickness	TO BE ADDED
Board	Length × width	TO BE ADDED
Element	Decagon circumradius	TO BE ADDED
Element	Feed-line width and slot width	TO BE ADDED
Element	Ground patch length × width	TO BE ADDED
Element	Spacing between elements	TO BE ADDED
FSS cell	Outer and inner ring radii	TO BE ADDED
FSS cell	Ring width and split width	TO BE ADDED
FSS	Cell period; board size	TO BE ADDED
Stack	Gap h, antenna to FSS	TO BE ADDED

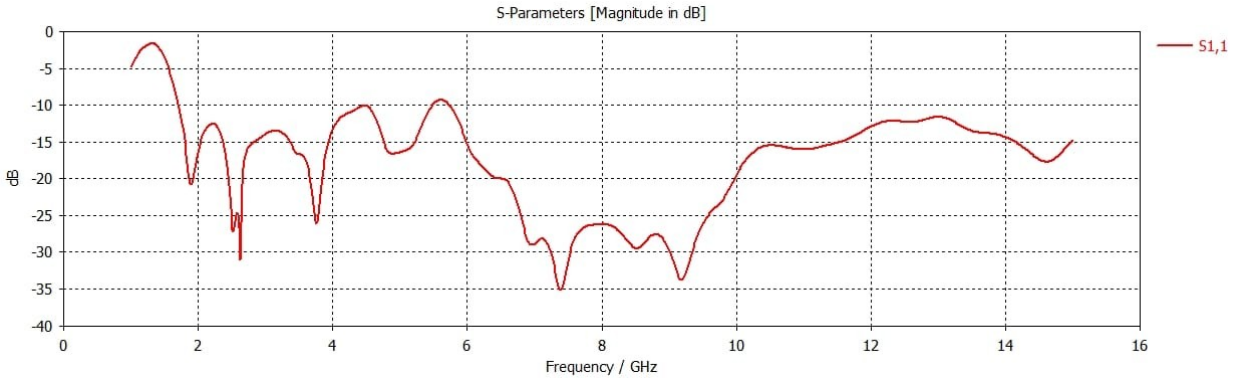


Fig. 4. Simulated reflection coefficient S_{11} at port 1 from 1 to 15 GHz (CST). TO BE CONFIRMED: taken here as the four-port antenna with the FSS.

III. RESULTS AND DISCUSSION

A. Reflection Coefficient

The structure was simulated in CST Studio Suite from 1 to 15 GHz. Fig. 4 shows the reflection coefficient at port 1, and Table II lists the values read from it. An antenna is taken as matched where S_{11} is below −10 dB, which means that less than 10 % of the input power is reflected.

S_{11} falls below −10 dB at 1.71 GHz and stays below it up to 5.48 GHz. This first band is 3.77 GHz wide, a fractional bandwidth of 105 %. From 5.48 to 5.73 GHz the curve rises slightly above the limit and reaches −9.2 dB at 5.62 GHz.

From 5.73 GHz it is below −10 dB again up to 15 GHz, the end of the simulated range, so the upper edge of this second band is not known yet. Over 5.73–15 GHz the fractional bandwidth is at least 89 %.

The curve has several resonances. The deepest is −34.9 dB at 7.38 GHz. Other resonances deeper than −20 dB are at 1.90, 2.62, 3.76, 8.50 and 9.18 GHz. The match is best from 6.5 to 10.0 GHz, where S_{11} stays below −20 dB. It is weakest near 5.62 GHz (−9.2 dB) and near 4.46 GHz (−10.1 dB). Above 10.5 GHz it stays between −11.5 and −17.6 dB. A small change of the feed gap or of the gap h could be tried to bring the region near 5.6 GHz below −10 dB, so that the two bands join into one.

TABLE II
REFLECTION COEFFICIENT AT PORT 1 (READ FROM FIG. 4)

Quantity	Value
Band 1, S_{11} < −10 dB	1.71–5.48 GHz (105 %)
Band 2, S_{11} < −10 dB	5.73–15 GHz† (≥ 89 %)
Highest S_{11} between the bands	−9.2 dB at 5.62 GHz
Deepest resonance	−34.9 dB at 7.38 GHz
Other resonances below −20 dB (GHz)	1.90, 2.62, 3.76, 8.50, 9.18

†End of the simulated range; the upper edge of band 2 is not known.

B. Results to Be Added

The reflection coefficient at one port is the only result available at present. The following results are needed to complete the paper and are not reported here: (i) S_{11} of the MIMO antenna without the FSS, for comparison; (ii) reflection coefficients at ports 2, 3 and 4; (iii) isolation between the ports (S_{21}, S_{31}, S_{41}); (iv) gain and efficiency against frequency, with and without the FSS; (v) ECC and diversity gain; (vi) radiation patterns: TO BE ADDED. Figs. 5–7 keep the place of the main missing plots. The ECC can be computed from the S-parameters with the method of [4] or from the far-field patterns [3].

TO BE ADDED
CST S-parameters of all four ports, 1–15 GHz

Fig. 5. Reflection coefficients of ports 2–4 and isolation between the ports. TO BE ADDED.

TO BE ADDED
CST realized gain (and efficiency) against frequency

Fig. 6. Gain of the MIMO antenna with and without the FSS. TO BE ADDED.

TO BE ADDED
CST far-field cuts in the two principal planes

Fig. 7. Radiation patterns at selected frequencies. TO BE ADDED.

C. Comparison with Reported Work

Table III places this work next to reported antennas that use an FSS or a metasurface reflector. The values of the other works are those given by their authors. The matched range of the present antenna is wide, but its gain and isolation must be added before a fair comparison can be made.

TABLE III
COMPARISON WITH REPORTED ANTENNAS

Ref.	Ports	Reflector	Band (GHz)	Peak gain (dBi)	Isolation (dB)
[14]	1	19 × 19 FSS	2.2–11.9	11.3	—
[15]	1	5 × 5 FSS	3–18	10.5	—
[18]	4	10 × 10 split-ring metasurface	3.08–7.75	8.3	> 15.5
This work	4	6 × 5 split-ring FSS	1.71–5.48, 5.73–15	TO BE ADDED	TO BE ADDED

—: single-port antenna. Values of the other works are peak gains reported by their authors.

IV. CONCLUSION

A four-port MIMO antenna with decagonal patches and CPW-style feeds was placed over an FSS made of a 6 × 5 array of concentric split rings, and the structure was simulated in CST Studio Suite. The simulated reflection coefficient at port 1 is below −10 dB from 1.71 to 5.48 GHz and from 5.73 to 15 GHz, with a short rise to −9.2 dB at 5.62 GHz between them, and the deepest resonance is −34.9 dB at 7.38 GHz. The antenna is therefore matched over most of the 2–15 GHz range. The dimensions, the isolation between the ports, the gain, the ECC and the radiation patterns are still to be added, and they will show how much the FSS improves the antenna. Fabrication and measurement of the antenna are planned as the next step.

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